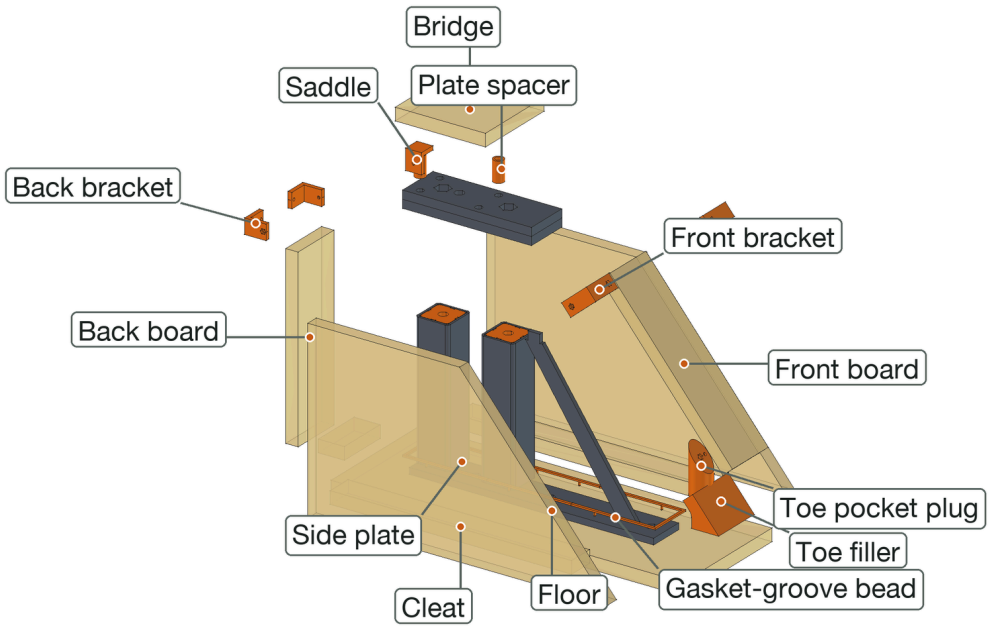
Upright molds

DIY CNC epoxy-granite mill · October 2026

How to make the upright molds. Drilling templates: `templates/upright-templates.pdf`. The complete guide (all the molds and the sheet layout for the boards) is `molds-complete.pdf`.

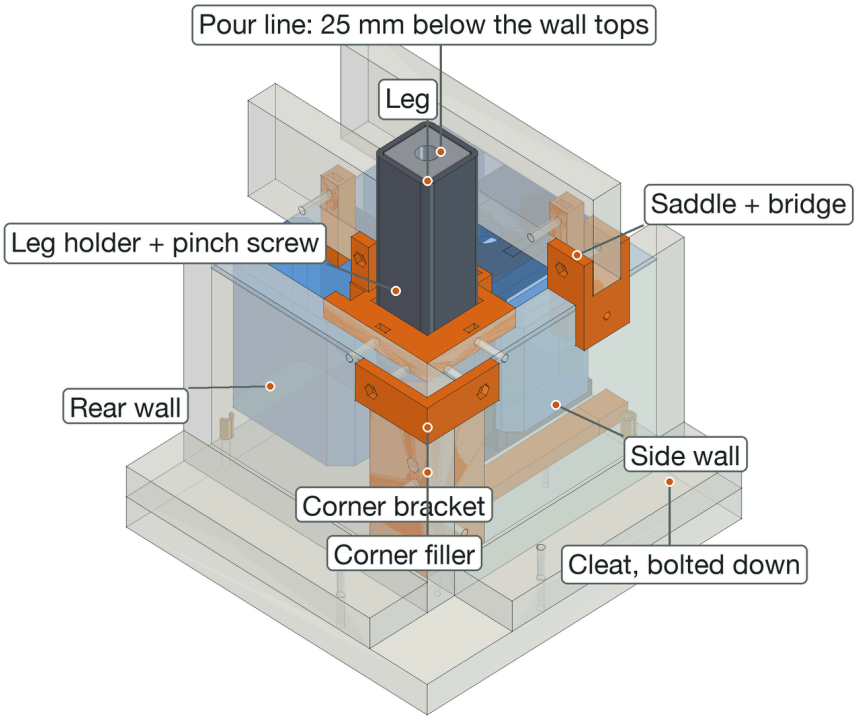


In short: melamine boards bolt together with printed brackets. Every printed piece bolts or pins to its board from outside. The walls stand 25 mm above the pour line, so no bolt goes through a face the mix touches. Nothing is glued, and it all comes apart for the next casting.

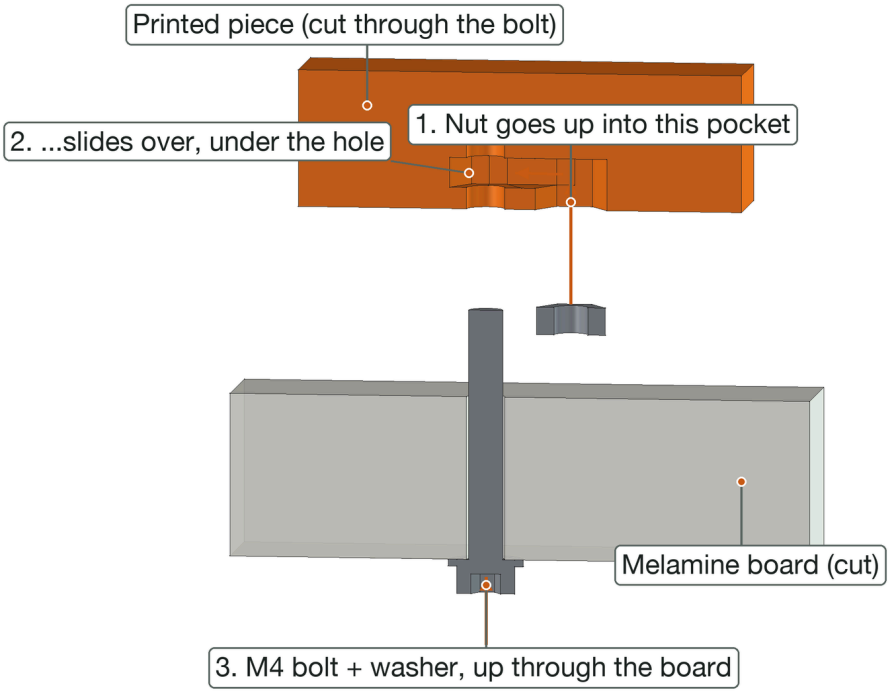
Order of work: print → cut the boards → mark them with the templates and drill → wax → assemble → pour → take apart.

1. How it works

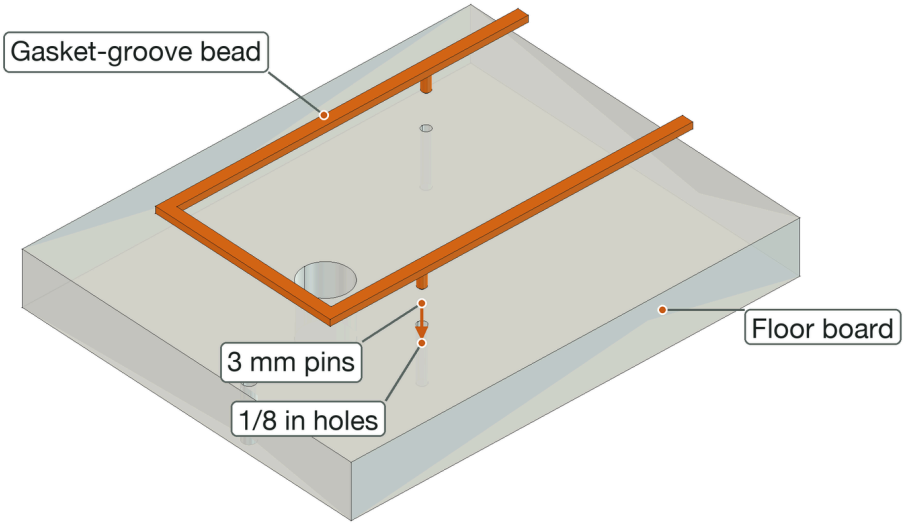
- **Walls** stand on the floor against bolted-down **cleats**; **corner brackets** square the tops. All the bolts go through the top 25 mm, above the pour line.



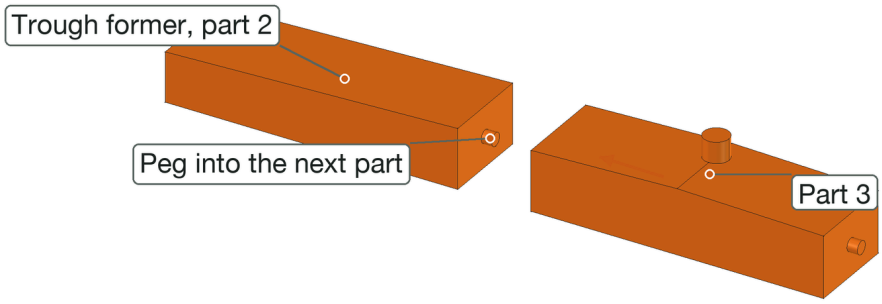
- **Printed pieces** have a nut built in: it goes into a pocket on the board side and slides under the hole; an M4 bolt comes through the board from outside.



- **Thin pieces** (gasket-groove beads, motor relief) have 3 mm pins that push into 1/8 in holes.



- **Long pieces** are printed in parts that peg together; each part bolts down on its own.



- **Fits** are 0.15 mm a side, and drawn for 19.05 mm board (Home Depot's 3/4 in melamine is a true 0.750 in). Measure yours; if it differs, set `mold_board` in `model/build_cnc.py` and re-run it.

2. What you need

Material	Amount	Notes
Melamine board, 3/4 in	about 0.9 m2	all the molds come out of two 4 x 8 ft sheets
PETG	0.32 kg	sliced; everything printed for the machine fits in two 1 kg rolls
Paste wax and PVA	1 each	release
3/8 in quarter-round PVC molding	a few meters	the rounded outside edges
2x4 offcut	4 blocks per upright mold, about 50 mm	feet

Hardware	Qty
M4 nut	40
M4 washer	40
M4 x 30 bolt	22
M4 x 35 bolt	4
M4 x 45 bolt	14
1/2 in rod (or a tie rod), greased, 355 mm: centers a top tie-rod sleeve	4
1/2-13 x 1-1/2 in bolt + nut, greased (forms a dowel hole)	4
3/8-16 x 3 in bolt, greased (holds the top plate, both layers)	4
5/8 in plain steel rod, greased (forms a slope tie-rod hole), cut to 89 mm	2

Add about 10 % spare small bolts, nuts and washers. **Tools:** drill and brad-point bits (sizes in section 5), a 2-1/2 in hole saw, a jigsaw for the side plates' slope, a square, clamps, a 7 mm nut driver, a 3 mm hex key, an awl.

3. Print

40 prints, 0.32 kg, about 23 hours (sliced; settings in the print guide). Print one corner bracket and one saddle first and check them on a board offcut. Press or slide the nuts in before the pieces go on the boards.

File	Qty	g	Time each
`upright/kit-back-bracket.stl`	4	21	27m 33s
`upright/kit-front-bracket.stl`	4	32	38m 26s
`upright/kit-saddle-flat.stl`	4	18	22m 40s
`upright/filler-toe-filler.stl`	2	51	1h 30m 59s
`upright/former-foot-gasket-groove-part1of4.stl`	2	6	16m 25s
`upright/former-foot-gasket-groove-part2of4.stl`	2	2	8m 22s
`upright/former-foot-gasket-groove-part3of4.stl`	2	2	8m 22s
`upright/former-foot-gasket-groove-part4of4.stl`	2	6	16m 30s

File	Qty	g	Time each
`upright/plug-toe-plug.stl`	2	26	1h 3m 42s
`upright/kit-plate-spacer.stl`	4	15	20m 9s
`tools/kit-pour-gauge.stl`	2	9	24m 33s
`tools/kit-screed-shoe.stl`	2	20	44m 42s
`upright/cast-in-column-ring-l1a.stl`	8	109	49m 49s

4. Cut the boards

Have the store rip the strips; crosscut at home. Write each board's name on its outside face, and mark the cast face (the melamine face the mix touches).

Board	Qty	Size (mm)	Notes
Upright floor	2	198 x 541	
Upright side plate, outer	2	461 x 271 (profile)	cut to the profile
Upright side plate, inner	2	461 x 271 (profile)	cut to the profile
Upright back board	2	80 x 271	
Upright front board	2	80 x 292	ends cut square
Upright bridge	2	131 x 138	
Upright cleat, outer	2	371 x 40	offcut strip, flat on the floor
Upright cleat, inner	2	371 x 40	offcut strip, flat on the floor
Upright cleat, back	2	80 x 40	offcut strip, flat on the floor

5. Mark and drill

1. Print the drilling templates (`templates/upright-templates.pdf`) at **actual size** and check the 100 mm square.
2. Tape each board's pages onto it: printed edges on the board's edges, grid lines lined up.
3. Awl through every crosshair. Pencil the dashed **pour line** on the walls.
4. Drill from the marked face, at the size the color legend gives. Pilot the big ones; the 63.5 mm holes are a 2-1/2 in hole saw.

The appendix has every board as a drawing and a table, if you would rather measure.

Hole	Drill	For
3.2 mm	1/8 in	foot-gasket-groove
4.5 mm	11/64 in	cleat, corner bracket, saddle, toe-filler, toe-plug
10 mm	25/64 in	3/8-16 bolt into the top plate
13.1 mm	33/64 in	1/2-13 bolt, base dowel hole
13.5 mm	17/32 in	1/2 rod through a sleeve
16.3 mm	41/64 in	5/8 core rod

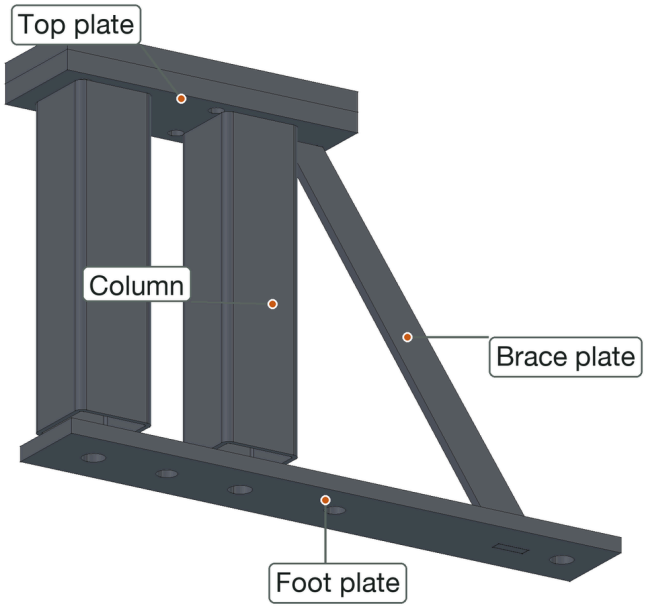
6. Wax

- 3-5 coats of paste wax on every face the mix touches, boards and removable printed pieces, then PVA. Don't wax pieces that stay in (core posts, cast-in parts): scuff and degrease them.
- Glue quarter-round into the floor-to-wall corners that make outside edges; silicone fillets elsewhere.
- Grease every screw, bolt and rod that comes out of the casting.

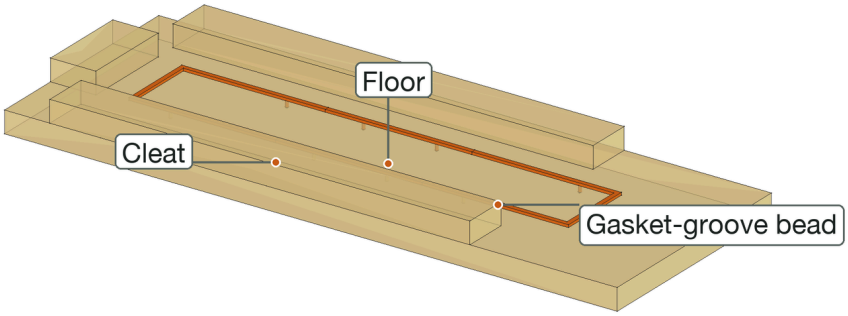
7. Assemble

Uprights (two molds)

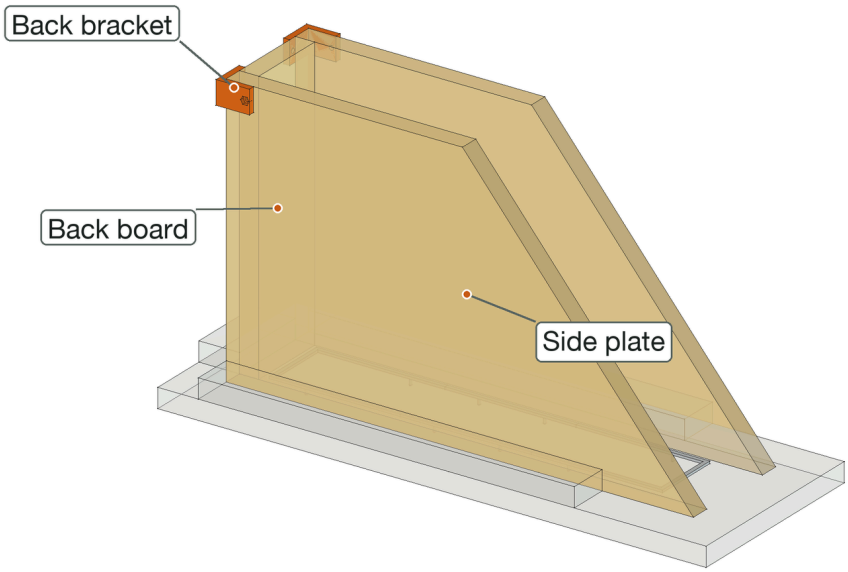
Cast foot down. Build two of everything. The skeleton: foot plate, two columns, the brace slotted into the foot plate, the top plate slotted onto the brace.



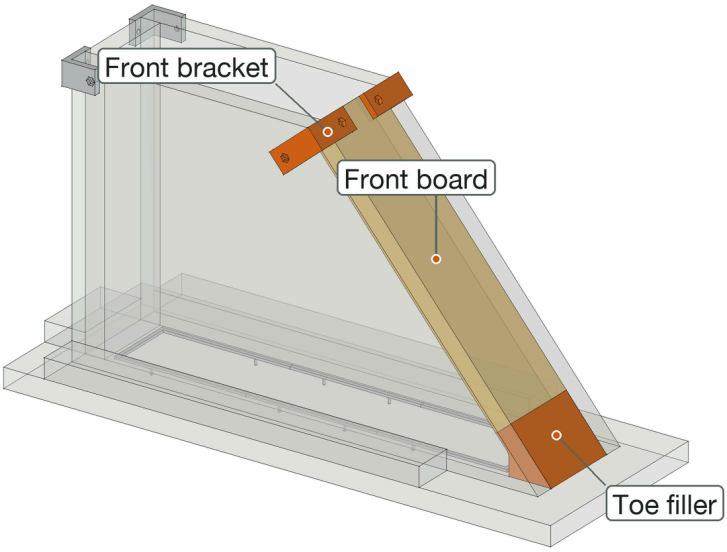
1. Gasket bead pinned on; cleats bolted down.



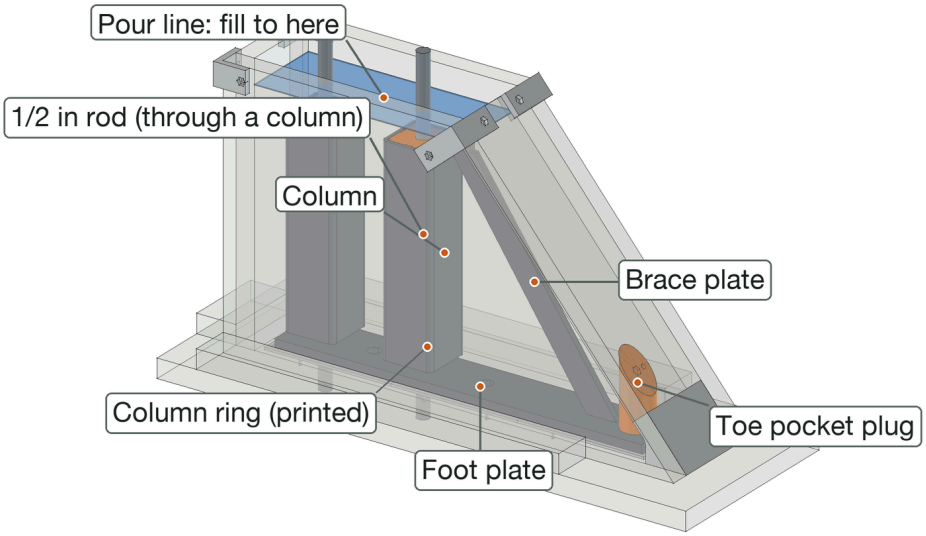
2. Side plates against their cleats, back board between them, two back brackets.



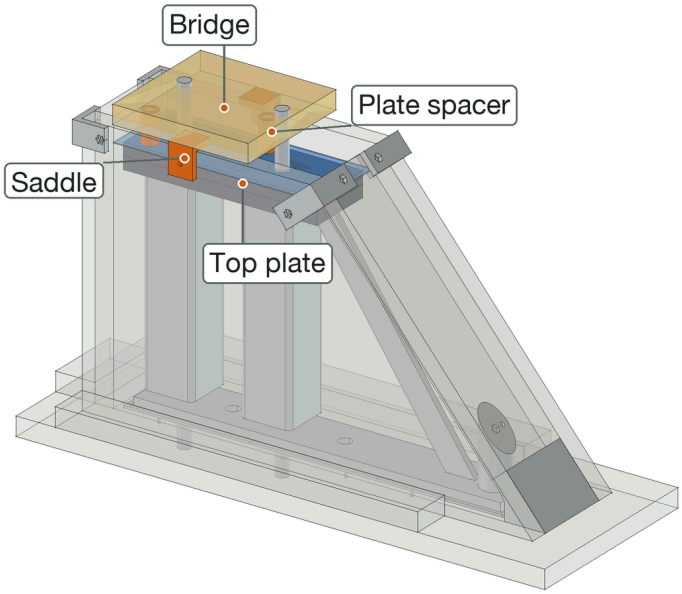
3. Toe filler bolted down, the front board's foot in its lip, two front brackets at the top.

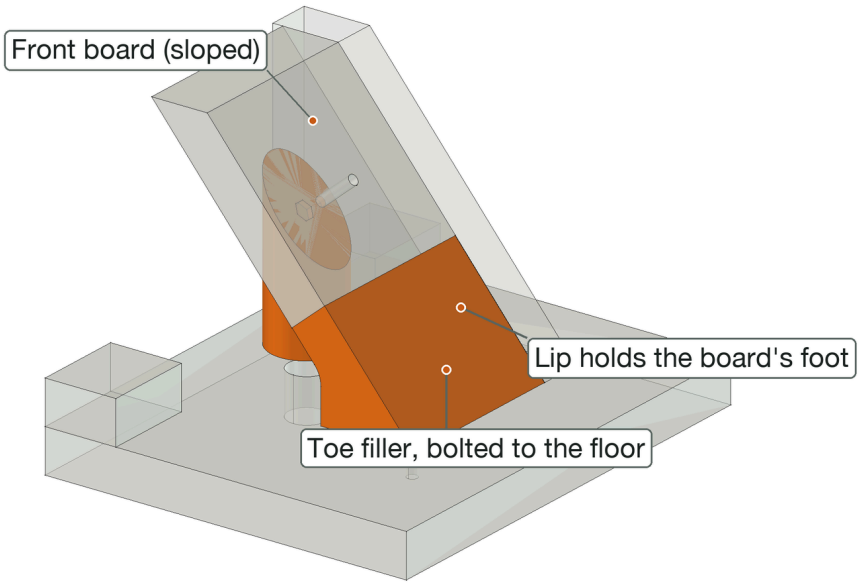


4. Mold on its four 2x4 blocks. The skeleton in: foot plate on the floor, greased 1/2 in rods up through the floor, the foot plate and the columns; the brace in its slot; the toe pocket plug on the foot plate; the two dowel-hole bolts from underneath. Columns on a bed of thickened epoxy, the top plate and bridge set on them while it goes off, then lifted away.



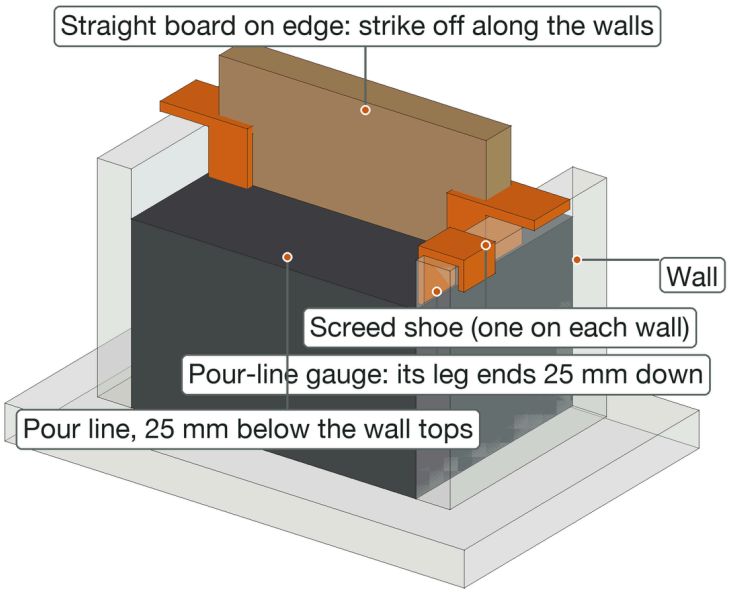
5. Fill to the line. Top plate (both layers) back down over the rods and the brace tab, hung from the bridge; bridge onto its saddles.





8. Pour

- Fill to the **pour line**, 25 mm below the wall tops (pencilled from the templates, or with the printed pour-line gauge clipped over a wall).
- Vibrate each layer until it stops bubbling; check the brackets halfway through.
- Strike the top off with a straight board in the two printed **screed shoes**: they hang it from the wall tops with its bottom edge on the pour line.



9. Take apart

1. After the full cure: bridges and saddles off, then corner brackets, then cleats; pull the walls off.
2. Screws, bolts and rods out from outside; lift the casting off the floor.
3. Unbolt and pull the formers and fillers.
4. Scrape and re-wax the boards; keep each mold's kit in a box. The kit is reused.

10. If something doesn't fit

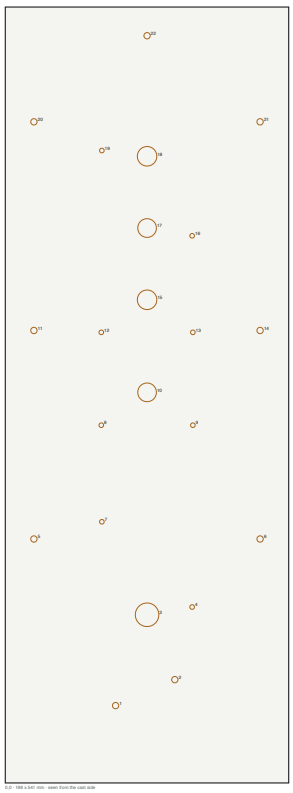
Problem	Fix
Nut won't slide under the hole	Clean the pocket with a nut on a bolt; warm the nut
Bracket or saddle loose on the board	Board thinner than set: tape shim, or set `mold_board` and reprint
Pin won't go in	Run the 1/8 in bit through again; trim the pin tip
Foam core moves	Tighten its post bolt; carpet tape under it

Problem	Fix
Printed part won't go on or into a steel tube	File the weld seam; sand the part. Several tight: caliper the tube, put its sizes in `frame_tube` / `post_tube`, re-run and reprint
Printed part loose on a tube	A wrap of tape round the part
Leak at a corner	Silicone fillet inside before waxing; check the brackets and diagonals
Former stuck after curing	Bolts out first, then a dowel and a light tap

Appendix: hole maps

Every board as seen from the face you mark: x from the left edge, y from the bottom edge (for a wall, the edge on the floor). The templates carry the same holes at full size.

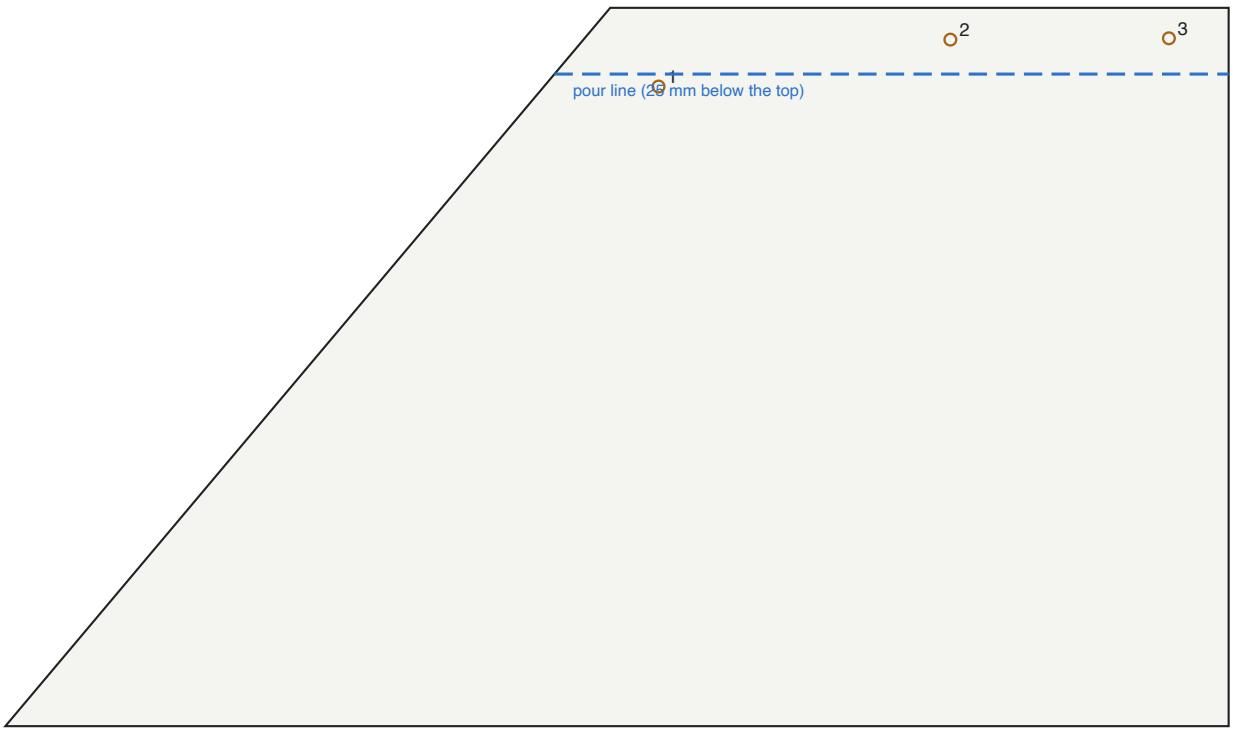
Upright floor (x2)



#	x	y	Hole	For
1	77	54	4.5	toe-filler
2	119	72	4.5	toe-filler
3	99	117	16.3	5/8 core rod
4	131	123	3.2	foot-gasket-groove
5	20	170	4.5	cleat
6	178	170	4.5	cleat
7	67	182	3.2	foot-gasket-groove

#	x	y	Hole	For
8	67	249	3.2	foot-gasket-groove
9	131	249	3.2	foot-gasket-groove
10	99	272	13.1	1/2-13 bolt, base dowel hole
11	20	315	4.5	cleat
12	67	314	3.2	foot-gasket-groove
13	131	314	3.2	foot-gasket-groove
14	178	315	4.5	cleat
15	99	337	13.5	1/2 rod through a sleeve
16	131	381	3.2	foot-gasket-groove
17	99	387	13.1	1/2-13 bolt, base dowel hole
18	99	437	13.5	1/2 rod through a sleeve
19	67	441	3.2	foot-gasket-groove
20	20	461	4.5	cleat
21	178	461	4.5	cleat
22	99	521	4.5	cleat

Upright side plate, outer (x2)



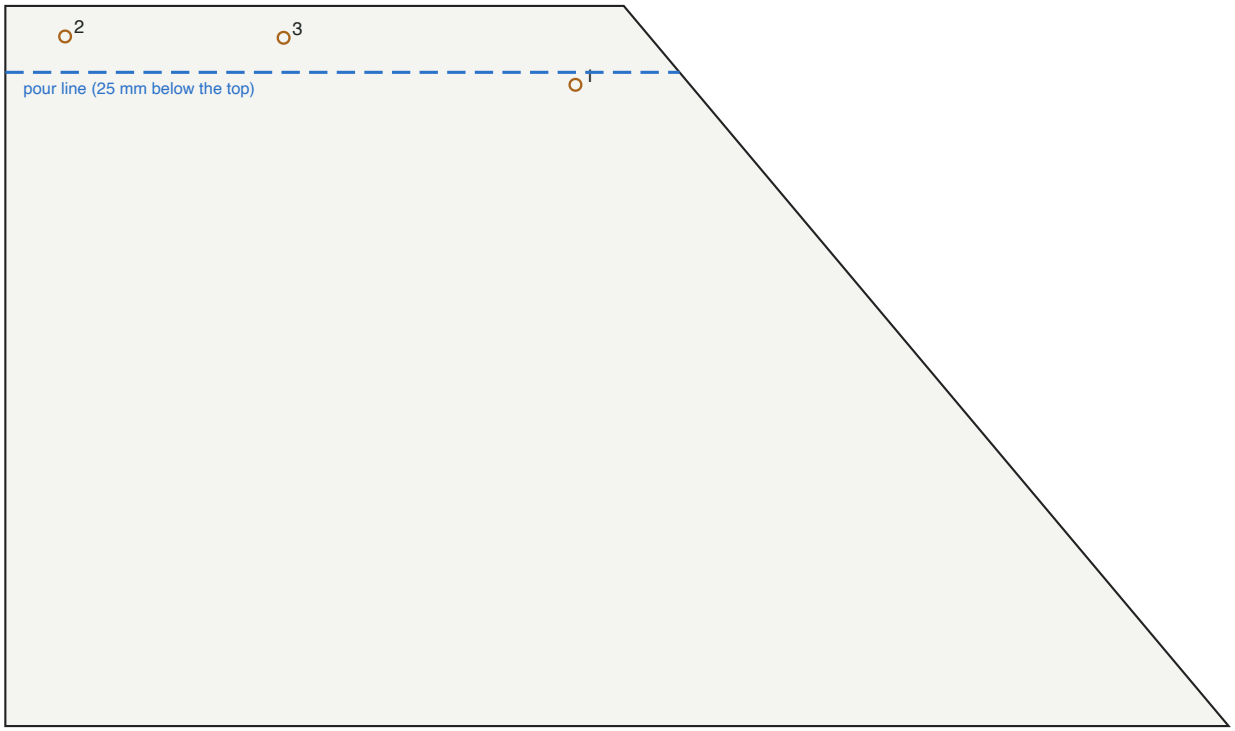
0,0 · 461 x 271 mm · seen from the cast side

#	x	y	Hole	For
1	246	241	4.5	corner bracket

#	x	y	Hole	For
2	356	259	4.5	saddle
3	438	260	4.5	corner bracket

Outline corners (x, y): (0, 0), (461, 0), (461, 271), (228, 271)

Upright side plate, inner (x2)

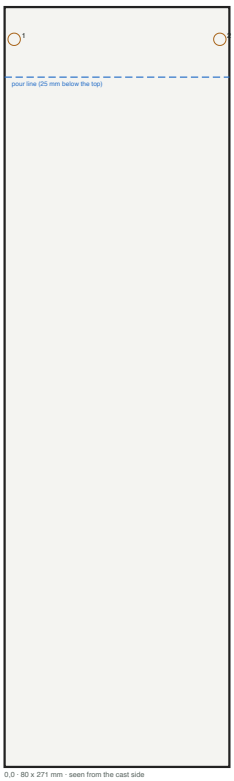


0,0 · 461 x 271 mm · seen from the cast side

#	x	y	Hole	For
1	215	241	4.5	corner bracket
2	22	260	4.5	corner bracket
3	105	259	4.5	saddle

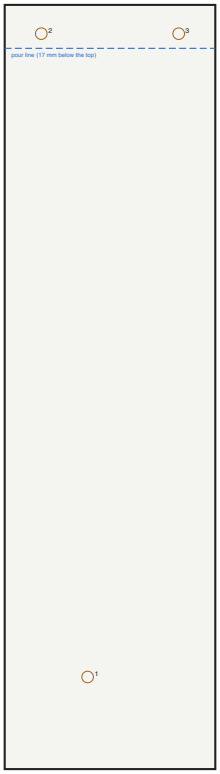
Outline corners (x, y): (461, 0), (0, 0), (0, 271), (233, 271)

Upright back board (x2)



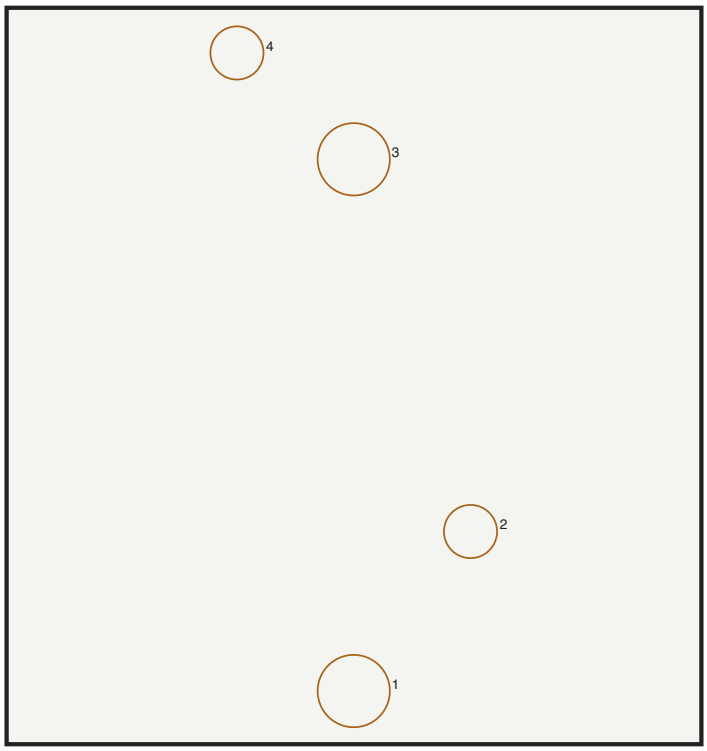
#	x	y	Hole	For
1	3	260	4.5	corner bracket
2	77	260	4.5	corner bracket

Upright front board (x2)



#	x	y	Hole	For
1	32	35	4.5	toe-plug
2	14	281	4.5	corner bracket
3	66	281	4.5	corner bracket

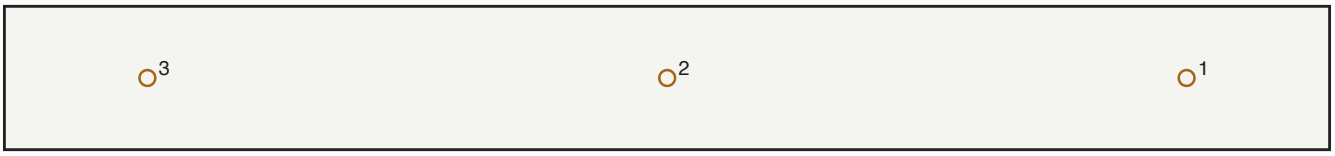
Upright bridge (x2)



0,0 · 131 x 138 mm · seen from the outside (top)

#	x	y	Hole	For
1	65	10	13.5	1/2 rod through a sleeve
2	87	40	10	3/8-16 bolt into the top plate
3	65	110	13.5	1/2 rod through a sleeve
4	43	130	10	3/8-16 bolt into the top plate

Upright cleat, outer (x2)



0,0 · 371 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	331	20	4.5	cleat
2	185	20	4.5	cleat
3	40	20	4.5	cleat

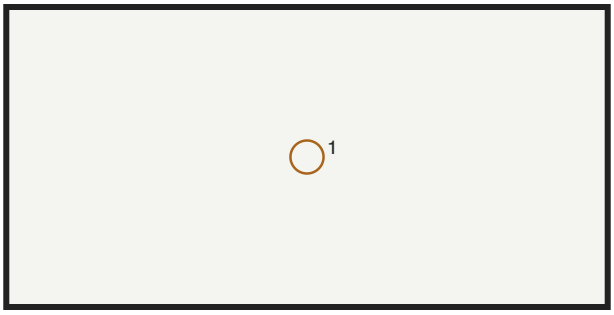
Upright cleat, inner (x2)



0,0 · 371 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	331	20	4.5	cleat
2	185	20	4.5	cleat
3	40	20	4.5	cleat

Upright cleat, back (x2)



0,0 · 80 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	40	20	4.5	cleat